Hot or Not
Technology 2012

We take a look at technologies that are just around the corner and conversely those that you didn't think were on their death bed

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As with life, change is the only constant in the tech world. Every year we keep coming across new technologies and at the same time, there a lot of technologies that slowly but surely ride into the sunset. The aim of this piece is to help you keep an eye out for technologies which are going to go mainstream in the coming years. Followed by those which we think are dying or already dead.

WHAT'S HOT
There's something in the air

Even though we live in an increasingly wireless world, there are still many wires around. Take a look at your desktop system and there is bunch of wires hidden somewhere. Somehow, transfer of electricity from the power source to the device still requires wires, with devices having batteries, you again require wires for charging. Sure, there is inductive charging, but that requires a close contact between the device and the source. So when do we get over wires? How about electricity going wireless?

The concept of wireless transfer of electricity was first worked on by Nikola Tesla back in the 1900s, but he wasn't entirely successful. Back in 2007, a group of researchers at MIT from the Department of Electronics and Computer Science and Department of Physics developed a prototype which transferred electricity wirelessly and lit a 60W bulb. They went on to form the company called WiTricity which is still conducting research on the practical applications of this technology.

Resonant energy transfer is the technology behind the experiment. It involves two resonant objects operating at resonant frequency. So the oscillating power emitter generates a magnetic field (as it is powered to a source), which is coupled with the receiver coil (which is oscillating at the same frequency as the emitter – resonant frequency). This causes the receiving coil to generate electricity which powers the devices it is connected to.

Resonant energy transfer taking place between the source emitter and receiver. The coupled magnetic field (in blue) causes the receiver coil to generate electricity which lights up the bulb

Conceptually it is similar to inductive stoves or inductive chargers, which use magnetic field to transmit electricity, but these devices work at a very small range. Resonant energy transfer allows you to extend that distance between the power emitter and the receiver by a few meters.

As this is a non-radiative form of energy transfer, involving coupling of the emitter and receiver using magnetic field, it is considered safe. Also the energy transfer will happen only if the receiver is resonating at the same frequency as the emitter.

Some of the applications include: direct wireless powering of devices in the home such as TV, microwave ovens, refridgerators, etc.; wireless charging of devices such as cellphones, tablets, laptops; industrial applications where you can wirelessly power or wirelessly charge key components on the factory floor; wirelessly charging electric cars.

Links to check out:
- WiTricity website http://goo.gl/xHrzL
- TED Video demoing wireless power transfer http://goo.gl/wBef2

Super high-res TV screens. Yeah we are talking 4K

As if the full HD resolution was not enough, TV manufacturers will soon come out with display panels which will take the resolution from 1920x1080 which we currently call full HD to 4096x2160 which will